

Limit

Limit of a function: When x approaches a constant quantity a from either side, if there exists a definite finite number l , towards which $f(x)$ approaches such that the numerical difference of $f(x)$ and l can be made as small as we please by taking x sufficiently close to a , then l is defined as the limit of $f(x)$ as x tends to a . This is symbolically written as $\lim_{x \rightarrow a} f(x) = l$.

P.1: A function $f(x)$ is defined as follows: $f(x) = \begin{cases} 2x-3, & 0 \leq x \leq 2 \\ x^2-3, & 2 < x \leq 4 \end{cases}$, Does

$\lim_{x \rightarrow 2} f(x)$ exist?

Solution:

$$\text{L.H.L} = \lim_{x \rightarrow 2^-} f(x)$$

$$= \lim_{h \rightarrow 0} f(2-h)$$

$$= \lim_{h \rightarrow 0} \{2(2-h)-3\}$$

$$= \lim_{h \rightarrow 0} (1-2h)$$

$$= 1-0$$

$$= 1$$

$$\text{R.H.L} = \lim_{x \rightarrow 2^+} f(x)$$

$$= \lim_{h \rightarrow 0} f(2+h)$$

$$= \lim_{h \rightarrow 0} \{(2+h)^2-3\}$$

$$= \lim_{h \rightarrow 0} (4+4h+h^2-3)$$

$$= \lim_{h \rightarrow 0} (1+4h+h^2)$$

$$= 1+0+0$$

$$= 1$$

Since L.H.L.=R.H.L, then $\lim_{x \rightarrow 2} f(x)$ exist

Continuity

Continuity of a function: Geometrically, If the graph of a function $y = f(x)$ is a continuous curve we naturally call the function a continuous one. It means there should not be any sudden change in the value of the function i.e. a small change in the value of x should produce a small change in the value of y and so the graph of the function should be a continuous curve without any break in it.

Lecture: 02
Limit, Continuity and Differentiability

P.1: Test the continuity of $f(x)$ at $x=0$, where $f(x) = \begin{cases} \frac{e^{\frac{1}{x}}}{e^{\frac{1}{x}} + 1}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$.

Solution:

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{h \rightarrow 0} f(0-h) \\ &= \lim_{h \rightarrow 0} \frac{e^{\frac{1}{0-h}}}{e^{\frac{1}{0-h}} + 1} \\ &= \lim_{h \rightarrow 0} \frac{e^{-\frac{1}{h}}}{e^{-\frac{1}{h}} + 1} \\ &= \frac{0}{0+1} \\ &= 0 \end{aligned}$$

and $f(0) = 0$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{h \rightarrow 0} f(0+h) \\ &= \lim_{h \rightarrow 0} \frac{e^{\frac{1}{0+h}}}{e^{\frac{1}{0+h}} + 1} \\ &= \lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}}}{e^{\frac{1}{h}} + 1} \\ &= \lim_{h \rightarrow 0} \frac{1}{1 + e^{-\frac{1}{h}}} \\ &= \frac{1}{1+0} \\ &= 1 \end{aligned}$$

Since L.H.L. $\neq$ R.H.L., the given function is discontinuous at $x=0$.

P.2: Test the continuity of $f(x)$ at $x=0$ where $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$.

Solution:

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{h \rightarrow 0} f(0-h) \\ &= \lim_{h \rightarrow 0} (0-h)^2 \sin \frac{1}{(0-h)} \\ &= - \lim_{h \rightarrow 0} h^2 \sin \frac{1}{h} \\ &= 0 \times [\text{a finite number between } -1 \text{ and } 1] \\ &= 0 \end{aligned}$$

and $f(0) = 0$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{h \rightarrow 0} f(0+h) \\ &= \lim_{h \rightarrow 0} (0+h)^2 \sin \frac{1}{(0+h)} \\ &= \lim_{h \rightarrow 0} h^2 \sin \frac{1}{h} \\ &= 0 \end{aligned}$$

Since L.H.L. = R.H.L. = $f(0)$, the given function is continuous at $x=0$.

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P.3: Test the continuity of $f(x)$ at $x=0$, where $f(x) = \begin{cases} \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1}, & \text{when } x \neq 0 \\ 0 & , \text{ when } x=0 \end{cases}$.

Solution:

$$\text{L.H.L} = \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{h \rightarrow 0} f(0-h)$$

$$= \lim_{h \rightarrow 0} \frac{e^{\frac{1}{0-h}} - 1}{e^{\frac{1}{0-h}} + 1}$$

$$= \lim_{h \rightarrow 0} \frac{e^{-\frac{1}{h}} - 1}{e^{-\frac{1}{h}} + 1}$$

$$= \frac{0 - 1}{0 + 1}$$

$$= -1$$

$$\text{R.H.L} = \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{h \rightarrow 0} f(0+h)$$

$$= \lim_{h \rightarrow 0} \frac{e^{\frac{1}{0+h}} - 1}{e^{\frac{1}{0+h}} + 1}$$

$$= \lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1}$$

$$= \lim_{h \rightarrow 0} \frac{1 - e^{-\frac{1}{h}}}{1 + e^{-\frac{1}{h}}}$$

$$= \frac{1 - 0}{1 + 0}$$

$$= 1$$

and $f(0) = 0$

Since L.H.L. $\neq$ R.H.L. $\neq f(0)$, the given function is discontinuous at $x=0$.

P.4: Find a nonzero value for the constant k that makes $f(x) = \begin{cases} \frac{\tan kx}{x}, & x < 0 \\ 3x + k^2, & x \geq 0 \end{cases}$ continuous

at $x = 0$.

Solution:

$$\text{L.H.L} = \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{h \rightarrow 0} f(0-h)$$

$$= \lim_{h \rightarrow 0} \frac{\tan k(0-h)}{(0-h)}$$

$$= \lim_{h \rightarrow 0} \frac{\tan kh}{h}$$

$$= \lim_{kh \rightarrow 0} \frac{\tan kh}{kh} \cdot k$$

$$= 1 \cdot k$$

$$= k$$

$$\text{R.H.L} = \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{h \rightarrow 0} f(0+h)$$

$$= \lim_{h \rightarrow 0} 3(0+h) + k^2$$

$$= k^2$$

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Limit, Continuity and Differentiability

$$\text{and } f(0) = 3.0 + k^2 \\ = k^2$$

Since the given function is continuous at $x=0$; L.H.L. = R.H.L. = $f(0)$

which gives $k^2 = k$

$$k(k-1) = 0$$

$$k = 0, 1$$

Therefore the nonzero value of the constant k is 1.

Differentiability

$$Lf'(a) = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h}$$

$$Rf'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

When $Lf'(a) = Rf'(a)$ and finite, then the given function is differentiable at $x=a$

P.1: Test the continuity and differentiability of $f(x) = \begin{cases} 1, & \text{when } x < 0 \\ 1 + \sin x, & \text{when } 0 \leq x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2, & \text{when } x \geq \frac{\pi}{2} \end{cases}$ **at**

$$x = \frac{\pi}{2} \text{ or } x = 0.$$

Solution: For $x = \frac{\pi}{2}$:

Continuity Test:

$$\text{Here } f\left(\frac{\pi}{2}\right) = 2 + \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^2 = 2$$

$$\text{L.H.L.} = \lim_{x \rightarrow \frac{\pi}{2}^-} f(x)$$

$$= \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} - h\right)$$

$$= \lim_{h \rightarrow 0} \left\{ 1 + \sin\left(\frac{\pi}{2} - h\right)^2 \right\}$$

$$\text{R.H.L.} = \lim_{x \rightarrow \frac{\pi}{2}^+} f(x)$$

$$= \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} + h\right)$$

$$= \lim_{h \rightarrow 0} 2 + \left(\frac{\pi}{2} + h - \frac{\pi}{2}\right)^2$$

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$$\begin{aligned} &= \lim_{h \rightarrow 0} (1 + \cos h) \\ &= 2 \end{aligned}$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} (2 + h^2) \\ &= 2 \end{aligned}$$

Since L.H.L. = R. H. L. = $f\left(\frac{\pi}{2}\right) = 2$, the given function is continuous at $x = \frac{\pi}{2}$.

Differentiability Test:

$$\begin{aligned} \text{Lf}'\left(\frac{\pi}{2}\right) &= \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2} - h\right) - f\left(\frac{\pi}{2}\right)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{1}{-h} \{ \cos h - 1 \} \\ &= \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \sin^2 \frac{h}{2}}{h} \\ &= \lim_{\frac{h}{2} \rightarrow 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}} \lim_{\frac{h}{2} \rightarrow 0} \sin \frac{h}{2} \\ &= 1 \times 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Rf}'\left(\frac{\pi}{2}\right) &= \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 + \left(\frac{\pi}{2} + h - \frac{\pi}{2}\right)^2 - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2}{h} \\ &= \lim_{h \rightarrow 0} h \\ &= 0 \end{aligned}$$

Since $\text{Lf}'(a) = \text{Rf}'(a) = 0$, the given function is differentiable at $x = \frac{\pi}{2}$.

For $x=0$:

Continuity Test:

Here $f(0) = 1 + \sin(0) = 1 + 0 = 1$

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} f(x)$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} f(0 - h) \\ &= \lim_{h \rightarrow 0} (0 - h)^2 \sin \frac{1}{(0 - h)} \\ &= - \lim_{h \rightarrow 0} h^2 \sin \frac{1}{h} \end{aligned}$$

$$\begin{aligned} &= 0 \times [a \text{ finite number between } -1 \text{ and } 1] \\ &= 0 \end{aligned}$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x)$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} f(0 + h) \\ &= \lim_{h \rightarrow 0} (0 + h)^2 \sin \frac{1}{(0 + h)} \\ &= \lim_{h \rightarrow 0} h^2 \sin \frac{1}{h} \end{aligned}$$

$$\begin{aligned} &= 0 \times [a \text{ finite number between } -1 \text{ and } 1] \\ &= 0 \end{aligned}$$

Since L.H.L. = R. H. L. $\neq f(0)$, the given function is not continuous at $x=0$.

Hence, the given function is not differentiable at $x=0$.

P.2: Test the continuity and differentiability of $f(x) = \begin{cases} x^2 + 1, & \text{when } x < 0 \\ x, & \text{when } 0 \leq x \leq 1 \\ \frac{1}{x}, & \text{when } x > 1 \end{cases}$ **at** $x=0$ **or** $x=1$.

P.3: Test the continuity and differentiability of $g(x) = \begin{cases} \ln x; & \text{when } 0 < x \leq 1 \\ 0; & \text{when } 1 < x \leq 2 \\ 1 + x^2; & \text{when } x > 2 \end{cases}$ **at** $x=2$

Solution: For $x=2$:

Continuity Test:

Here $f(2) = 0$.

$$\begin{aligned} \text{L.H.L.} &= \lim_{x \rightarrow 2^-} g(x) \\ &= \lim_{h \rightarrow 0} g(2-h) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 2^+} g(x) \\ &= \lim_{h \rightarrow 0} g(2+h) \\ &= \lim_{h \rightarrow 0} \{1 + (2+h)^2\} \\ &= \lim_{h \rightarrow 0} \{1 + 4 + 4h + h^2\} \\ &= \lim_{h \rightarrow 0} (5 + 4h + h^2) \\ &= 5 + 8 + 4 \\ &= 17 \end{aligned}$$

Since $L.H.S \neq R.H.S = f(2)$, the given function is not continuous at $x=2$.

Hence, the given function is not differentiable at $x=2$.